Montana Railroad History — GeoJSON Mapping Reference

This document describes the geo-mapping dataset produced for the 18-article Montana Railroad History series. It is intended as a reference guide for a coding agent building an interactive map.

Files in This Package

File	Format	Contents
`montana_railroad_map.geojson`	GeoJSON (RFC 7946)	All 42 features: 33 Points + 9 LineStrings
`montana_railroad_locations.csv`	CSV	33 Point features with lat/lon, metadata, article links
`montana_railroad_routes.csv`	CSV	9 LineString routes with waypoints, railroad, status

GeoJSON Structure

The GeoJSON file is a standard `FeatureCollection`. Every feature carries a `properties` object with the following fields:

Common Properties (all features)

Property	Type	Description
`id`	string	Unique slug identifier for the feature
`name`	string	Human-readable location or route name
`type`	string	`"point_of_interest"` or `"route"`
`subtype`	string	Detailed classification (see table below)
`article`	string	Full title of the linked article
`article_file`	string	Filename of the linked article (e.g. `bear-creek-coal-trains.md`)
`author_voice`	string	Author voice used for the article
`description`	string	1-3 sentence narrative description for map popups
`marker_color`	string	Hex color code for map rendering (per article/theme)

Point-Only Properties

Property	Type	Description
`year`	string	Year of primary historical significance
`county`	string	Montana county (or state for cross-border features)

Route-Only Properties

Property	Type	Description
`railroad`	string	Full railroad company name
`status`	string	`active` , `active_as_bnsf` , `abandoned` , `abandoned_trail`

Feature Subtypes

Subtype	Count	Description
`depot`	9	Historic or active railroad stations
`main_line`	3	Primary transcontinental route corridors
`railroad_town`	3	Towns platted or founded by railroads
`historic_site`	4	General sites of historical significance
`mine_disaster`	2	Sites of coal or hard rock mine disasters
`branch_line`	2	Secondary or industrial branch lines
`industrial_site`	2	Smelters and industrial facilities
`tunnel_portal`	2	Railroad tunnel portals
`disaster_site`	1	Train wreck site
`historic_ceremony`	1	Golden spike / completion ceremony site
`mountain_pass`	1	Continental Divide crossing
`monument`	1	Historic monument
`electrification_terminal`	1	End-of-electrification changeover point
`electrified_main_line`	1	Electrified railroad corridor
`historic_community`	1	Ethnic/immigrant community site
`indigenous_site`	1	Native American nation territory site
`abandonment_site`	1	Railroad abandonment marker point
`abandoned_right_of_way`	1	Abandoned railroad corridor
`historic_landing`	1	Missouri River steamboat landing
`passenger_main_line`	1	Active passenger rail corridor

Subtype	Count	Description
`labor_history`	1	Labor strike / riot site
`historic_district`	1	Designated historic district

Color Coding by Theme

Each article cluster uses a consistent hex color for marker rendering:

Color	Hex	Articles
Dark Red	`#8B0000`	Bear Creek Coal, Blackfeet Displacement, Butte Labor Riots
Navy Blue	`#1A3A5C`	Custer Creek Wreck, Marias Pass, Milwaukee Road Electrification
Gold	`#8B6914`	Gold Creek Ceremony, Chinese Workers, Fort Benton Steamboats
Purple	`#4A0080`	BA&P Railway, Copper Kings
Forest Green	`#2E5E1E`	Jawbone Railroad, Homestead Promoters, NP Towns
Saddle Brown	`#8B4513`	Milwaukee Road Electrification Route, St. Paul Pass Tunnel
Dark Green	`#1A5C2E`	Glacier Park, Empire Builder
Gray	`#555555`	Abandoned Milwaukee Road

Article-to-Feature Mapping

Article File	Points	Routes	Key Locations
`bear-creek-coal-trains.md`	2	1	Smith Mine, Red Lodge Depot
`custer-creek-train-wreck.md`	1	1	Custer Creek Wreck Site
`gold-creek-golden-spike.md`	1	1	Gold Creek Completion Site
`butte-anaconda-pacific-railway.md`	2	1	Butte Terminus, Anaconda Smelter
`jawbone-railroad.md`	2	1	Harlowton, Lewistown
`marias-pass-great-northern.md`	2	0	Marias Pass Summit, Stevens Monument
`milwaukee-road-electrification.md`	1	1	Harlowton Electrification Terminal
`homestead-boom-railroad-promoters.md`	2	0	Billings NP Land Office, Livingston Depot
`chinese-railroad-workers.md`	2	0	Helena Chinatown, Missoula NP Division
`blackfeet-displacement-railroad.md`	2	0	Browning, Cut Bank
`glacier-park-great-northern.md`	2	1	East Glacier Station, Whitefish Station
`st-paul-pass-tunnel.md`	2	0	East Portal (MT), West Portal (ID)
`abandoned-milwaukee-road.md`	1	1	Harlowton Abandonment Point
`end-of-steamboat-era.md`	2	0	Fort Benton Levee, GN Depot
`empire-builder-passenger-train.md`	1	1	Whitefish Empire Builder Station

Article File	Points	Routes	Key Locations
`butte-labor-riots-1914.md`	2	0	Miners Union Hall, Granite Mountain Mine
`northern-pacific-birth-of-towns.md`	3	0	Billings, Miles City, Glendive
`copper-kings-railroad-wars.md`	3	0	Butte Mine District, Anaconda, Great Falls

Route Status Values

Status	Meaning
`active`	Currently operating railroad
`active_as_bnsf`	Former NP/GN line, now operated by BNSF Railway
`abandoned`	Tracks removed, right-of-way may be visible
`abandoned_trail`	Right-of-way converted to recreational trail

Suggested Coding Agent Usage

Leaflet.js / Mapbox GL JS: Load `montana_railroad_map.geojson` directly. Use `feature.properties.type` to switch between Point (circle/icon markers) and LineString (polyline) rendering. Use `feature.properties.marker_color` for consistent color coding. Bind `feature.properties.description` and `feature.properties.article` to popup content.

Python (GeoPandas / Folium):

```
import geopandas as gpd
gdf = gpd.read_file('montana_railroad_map.geojson')
points = gdf[gdf.geometry.type == 'Point']
routes = gdf[gdf.geometry.type == 'LineString']
```

Filtering by article:

```
article_features = gdf[gdf['article_file'] == 'bear-creek-coal-trains.md']
```

CSV for tabular tools (ArcGIS, QGIS, Google Maps import): Use `montana_railroad_locations.csv` for point import. Latitude column: `latitude`, Longitude column: `longitude`.